

# Immigrant workforce and agriculture productivity: evidence from Italian farm-level data

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## Abstract

This paper investigates the farm-level relationship between total factor productivity (TFP) and presence and composition of immigrant workers. A theoretical model expressing how immigrant workforce influences productivity is developed and an explicit and consistent representation of the so-called ‘diversity hypothesis’ is derived. A dynamic panel specification is estimated, and the respective econometric issues are discussed. The analysis is applied to a balanced panel of Italian professional farms extracted from the Farm Accountancy Data Network over the period 2008–2015. The use of micro-level data in such assessment allows to take the wide heterogeneity across farms into account. Results suggest that, when productivity dynamics and farm heterogeneity are accounted for, empirical support to the diversity hypothesis can be found in Italian professional agriculture.

**Key words:** immigrant workforce, diversity, total factor productivity, farm-level data, dynamic panel models

**JEL classification:** Q12, D24, J61

## 1. Introduction

This paper investigates the impact of immigrant workers (UN, 1990) on farm-level productivity performance. This issue is of undisputable political relevance, but, as evidenced by the body of recent literature on the subject, it is also theoretically and empirically challenging. In fact, there are several ways through which immigration may affect productivity, and these effects may be complex and not clear-cut.

The present study contributes to the literature as follows. Firstly, based on the recent literature, it puts forward a modelling framework that explains the complex impact of immigrant workers on farm-level productivity. This model

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leads to a testable specification of the relationship, commonly referred to as the ‘diversity hypothesis’. Secondly, the paper proposes an empirical strategy to assess this hypothesis and presents an application using Italian farm-level data. Italian agriculture is a suitable case study, given its wide heterogeneity of farming systems, productive performance and presence of immigrants.

The paper is structured as follows. Section 2 summarises the relevant theoretical and empirical literature on the major channels through which immigrant workers can affect productivity focusing on the diversity hypothesis and its empirical assessment. Section 3 presents the modelling approach to assess the farm-level relationship between productivity and presence and composition of immigrant workforce. Econometric implications and issues are also examined. Section 4 discusses the Italian case study, the data set and the methodology used to compute the farm-level total factor productivity (TFP) indexes. Section 5 presents and discusses the estimation results, while Section 6 draws some policy implications and concludes.

## 2. Overview of the literature and main research questions

### 2.1. The economics of diversity

The present study addresses a specific channel through which migration affects productivity. It focuses on immigrant workforce and productivity within a firm under constant wages, so it does not extend to productivity performance through the labour market and the economy sectoral composition.<sup>1</sup> It will here be named the ‘diversity effect’, following the significant body of recent theoretical and empirical literature on the possibly conflicting mechanisms that lead to this micro-level effect (Alesina and La Ferrara, 2005; Alesina, Harnoss and Rapoport, 2018).

Diversity, in terms of different characteristics of the workforce, may affect productivity both positively and negatively. The positive effect on productivity can derive from production complementarities that would not be obtained under homogeneous production factors. In this respect, even the presence of low-skilled workers may have a positive impact on firm-level productivity due to its mix with high-skilled ones (Bandiera, Barankay and Rasul, 2009; Gomez-Tello and Nicolini, 2017). This mix, in turn, may induce more appropriate managerial practices, learning processes and positive social spillovers stemming from co-working (Boom and van Reenen, 2007; Bandiera, Barankay and Rasul, 2009; Caliendo and Rossi-Hansberg, 2012; Gómez-Tello and Nicolini, 2017). The interaction among workers with different skills and cultural backgrounds may also bring new knowledge and

<sup>1</sup> The other two effects of immigration on productivity can be referred to as ‘labour-market effect’ and the ‘composition effect’. The former occurs when low(high)-skilled immigrants induce lower (higher) wages and, thus, lower (higher) labour productivity (Garcia-Alvarez-Coque, 2002; Dustmann, Glitz and Frattini, 2008). The latter occurs whenever low(high)-skilled immigrant workers selectively favour less (more) productive sectors, thus changing the composition of the economy and producing a negative (positive) impact on the economy-wide productivity (Kangasniemi et al., 2012).

ideas and, thus, favour innovation, which eventually increases productivity (Malchow-Møller *et al.*, 2013; Parrotta, Pozzoli and Pytlikova, 2014; Fassio, Montobbio and Venturini, 2015; Ottaviano, Peri and Wright, 2018; Bratti and Conti, 2018; Boberg-Fazlic and Sharp, 2019). This complementarity can also be external to the individual production unit and emerge at a more aggregate level through knowledge spillovers generated by the higher mobility of immigrant workers (Faggian and McCann, 2009; Simonen and McCann, 2010).

On the other hand, diversity can affect growth and productivity in a negative way due to additional costs known as ‘costs of fractionalisation’ (Easterly and Levine, 1997; Alesina *et al.*, 2003; Posner, 2004). Differences in culture and language can raise communication costs and increase conflicts, or reduce trust, cooperation and social capital (Alesina and La Ferrara, 2005). These workplace consequences of migration are found to depend only marginally on the education level and skilfulness of immigrants, though a negative diversity effect is more likely to occur with low-skilled workforce.

The overall result emerging from the literature is that the productivity gain of interaction decreases with social distance among workers and that it may become negative. So, there could be an optimal degree of diversity of the workforce. This diversity hypothesis has an interesting empirical implication. If diversity were measurable, a non-monotonic relationship between the diversity measure and productivity should be observed with diversity initially bringing productivity improvement after which it becomes ‘too much’, leading to a decrease of productivity. This suggests an inverted-U-shape relationship with the turning point representing the above-mentioned optimality (Ashraf and Galor, 2013).

The recent theoretical contribution by Gradstein and Justman (2019) presents this diversity hypothesis as a labour-augmenting factor within a production function framework. This factor depends on the interactions among workers and increases productivity if the social, cultural and ethnic distance is small. The effect of interaction decreases with this distance, and, beyond some point, interaction among workers may generate a negative impact on productivity.

## 2.2. The empirics of diversity

Empirical assessment of the diversity hypothesis is not easy for two main reasons. The first empirical issue is with isolating the ‘diversity effect’ from the labour market and composition effects. When working with macro-level or aggregate (country, regional, sectoral) data, the three effects coexist and are hardly distinguishable (Hanson and Slaughter, 2002; Dustmann, Glitz and Frattini, 2008). Gómez-Tello and Nicolini (2017) stress that the composition effect at the aggregate level may downsize, offset and even revert the effect occurring within the firms of any specific sector. The diversity effect can thus be separately identified only by using micro-level (firm-/farm-level) data (Malchow-Møller *et al.*, 2013). Assessing the relationship between the presence of immigrant workers and productivity with micro-level data of a specific

sector eliminates the sectoral composition effect. Moreover, cross-firm comparisons allow to mimic the constant-wage and constant-sector conditions by controlling for possible wage and production differentials.

In this respect, agriculture seems an interesting sector given the large heterogeneity in terms of the presence of immigrant workers as well as in productive performance across farm production specialisations, size, factor endowment and geographical location. Looking for the diversity effect within agriculture implies assessing the farm-level relationship between immigrant workforce and productivity controlling for the confounding factors, such as specialisation, size and location. Taking these aspects into account requires the use of farm-level data (Gómez-Tello and Nicolini, 2017; Malchow-Møller *et al.*, 2013).<sup>2</sup> In the agricultural economics literature, however, most of the empirical evidence on the impact of immigrant workers comes from aggregate data (Devadoss and Luckstead, 2008; Partridge, Rickman and Ali, 2008). Two exceptions are Malchow-Møller *et al.* (2013) who present a micro-level analysis of Danish farms over the 1980–2008 period and Baldoni, Coderoni and Esposti (2017) who focus on a panel of Italian farms over the 2008–2015 period. Both studies report a small, positive correlation between immigrant employment and farm revenues, but no significance of a positive impact on labour productivity.

The second issue of the empirics of diversity is that diversity of the workforce, by itself, is an abstract concept. According to Kemeny (2017: 169), ‘immigrant diversity remains a latent concept, and thus not something that can be uniquely or precisely identified’. Diversity must be represented and measured as a composite indicator combining observable characteristics of presence and composition of the firm/farm immigrant workforce. This composite indicator should be flexible enough to allow for a non-monotonic relationship with the farm-level productivity and to capture the possibly complex influence and interaction of the single observable characteristics. If properly constructed, this composite indicator of diversity would allow to disentangle the two possible directions within the immigrants–productivity relationship implied by the diversity hypothesis discussed above.

2 In fact, a farm-level analysis can point out that also a ‘within-sector composition effect’ can occur. The association of immigrant workers and productivity can emerge only because immigrants tend to be more present in farms showing a higher (or lower) productivity in any case. It could also be argued that the observed relationship might remain spurious due to the different factor endowment across farms. Higher (lower) productivity can be associated with immigrant workers just because they are more present in farms with higher (lower) factor endowment in terms of capital, materials and other labour force. However, if the assessment is made on the TFP and not on partial factor productivity, the role of immigrants should be considered independent of factor endowment. An investigation of the impact of immigrant workforce on farm-level labour productivity can be found in Baldoni, Coderoni and Esposti (2017).

### 3. The model

#### 3.1. The theoretical background

Consider a panel of  $N$  farms observed over  $T$  periods and represent the farm-specific production technology with the following generic neoclassical production function (Chambers, 1988):

$$Y_{it} = F^i(K_{it}, L_{it}^*, t), \quad i = 1, \dots, N; t = 1, \dots, T \quad (1)$$

where  $Y$  represents the aggregate farm output,  $L^*$  and  $K$  are labour and capital inputs (with  $K$  aggregating, for simplicity, all the non-labour production factors), and  $t$  is the usual time trend proxying the unobserved level of the technology.  $L^*$  measures the ‘effective’ labour, that is, the actual labour corrected by a labour-augmenting term. Considering the focus of the present study, this labour-augmenting term is here related to labour force diversity. Therefore, this ‘effective’ labour can be represented as  $L_{it}^* = D_{it}L_{it}$ , where  $D_{it}$  expresses the degree of diversity that augments labour productivity, hereafter ‘productive diversity’.

The assumption is that neither the analyst nor the farmer can directly observe diversity, but the latter can observe the impact of diversity on farm performance.<sup>3</sup> Therefore, as diversity affects labour productivity, it affects the farmer’s input use decisions.<sup>4</sup>

Equation (1) includes two sources of TFP growth. To make this more explicit, equation (1) can be rewritten as follows (Hulten, 1992):

$$Y_{it} = A_t F^i(K_{it}, D_{it}L_{it}), \quad \text{where } A_t = A_0 e^{\lambda t} \quad (2)$$

where  $A_t$  and  $A_0$  indicate the technological level at time  $t$  and in the initial period, respectively.  $\lambda$  is the conventional exogenous Hicks-neutral technological change rate. By differentiating equation (2) over time (Hulten, 1992), we obtain the following expression:

$$y_{it} - (\theta_{kit}k_{it} - \theta_{Lit}l_{it}) = \lambda + \theta_{Lit}d_{it} \quad (3)$$

where  $y_{it}, k_{it}, l_{it}, d_{it}$  express the growth rates of  $Y, K, L$  and  $D$ , respectively, and  $\theta_{kit}$  and  $\theta_{Lit}$  denote output elasticities of capital ( $K$ ) and labour ( $L$ ). Under profit maximisation, competitive input markets and constant returns to scale,  $\theta_{kit}$  and  $\theta_{Lit}$  correspond to the respective factor shares.<sup>5</sup> Equation (3) clarifies that, within the adopted framework, TFP growth is a combination

3 Diversity is here assumed to be a free attribute of the labour force. Consequently, it does not affect wage (i.e. all workers receive the same wage).

4 For a discussion on the similarity of this representation with previous models embodying technological change (for instance, through learning-by-doing), see Esposti (2011).

5 The assumption of constant returns to scale is not strictly needed within the adopted theoretical framework. However, the index number TFP calculation uses factor shares and thus implicitly assumes constant returns to scale. Therefore, under non-constant returns to scale, the TFP calculation performed with index numbers also incorporates the impact of scale economies on productivity. The empirical implication of this issue is clarified later.

of two terms: the Hicks-neutral exogenous technological change ( $\lambda$ ) and the labour-augmenting effect of productive diversity ( $\theta_{Lit}d_{it}$ ), behaving as a Harrod-neutral technological change.

### 3.2. The empirical strategy

The identification of the two terms on the right-hand side of equation (3) requires the identification and measurement of its left-hand term. This can be obtained following two different empirical strategies (Antle and Capalbo, 1989: 62–63). The first is the specification and estimation of equation (2) as a fully stochastic flexible production/cost/profit/distance function, while the second strategy is based on the growth accounting approach and index number theory (Coelli *et al.*, 2005; Fried, Lovell and Schmidt, 2008). The theoretical equivalence of the two approaches, once appropriate indexes are adopted,<sup>6</sup> has been clearly demonstrated (Good, Nadiri and Sickles, 1999; OECD, 2001), but the former approach has become the standard in academic research in this field.

The econometric approach has advantages compared with the use of index numbers. Firstly, the TFP calculation comes from a stochastic representation of the production technology, thus allowing to account for all the stochastic noise (i.e. all possible errors, deviations and mismeasurements) associated with the relationship between inputs and outputs and, thus, with TFP measurement. Secondly, TFP measurement emerges from an explicit and flexible representation of the production technology. This allows to properly consider the substitution/complementary effects among inputs and among outputs; for instance, those between native and immigrant workers and between these two workforces and capital. The explicit and flexible representation of the technology leads to a third advantage consisting of the estimation of both TFP and its explanation or decomposition, through a proper inclusion of TFP determinants or the specification of some characteristics of the technology (for instance, non-constant returns to scale) (O'Donnell, 2016; Njuki, Bravo-Ureta and O'Donnell, 2019).

But with these advantages also come disadvantages related to the unobserved heterogeneity in the data set. The econometric approach assumes a common technology across all observations; failing to fully account for the heterogeneity may induce an incorrect estimation of the technology and, consequently, an incorrect and misleading TFP measurement (Greene, 2005; Lachaud, Bravo-Ureta and Ludena, 2017; Njuki, Bravo-Ureta and O'Donnell, 2019: 840). Thus, a misspecification of the technology generates a deterministic noise in the estimation stage.<sup>7</sup> This issue is particularly relevant when micro-level data are used and even more so for the agricultural sector

6 Diewert introduced the concept of *superlative indexes* to designate indexes that are discrete-time approximations of the Divisia indexes and are exact for flexible functional forms (Diewert, 1992; OECD, 2001: 88).

7 According to Greene (2005: 270), whenever 'the true underlying technology contains unmeasured firm specific characteristics that reflect the technology in use [...] the model estimated by the analyst is actually incomplete or misspecified'.

where external environmental factors may amplify the degree of unobserved heterogeneity (Mundlak, 1989; Esposti, 2017).

Solutions to this issue put forward in the recent literature are methodologically interesting but empirically challenging.<sup>8</sup> Maintaining a flexible representation of the technology and admitting all the possible sources of heterogeneity across units implies complex empirical specifications, data demanding estimation and brings about complex econometric issues demanding estimation and brings about complex econometric issues. Moreover, the stochastic nature of these approaches only apparently takes all the possible stochastic noise into account. In fact, they commonly require strict distributional assumptions that exclude specific (and often extreme) farm-level errors and deviations frequently occurring in micro-level panels, thus becoming themselves sources of incorrect TFP estimation (O'Donnell, 2016).

More importantly, these solutions do not properly account for one of the major sources of heterogeneity across farms, that is, the different production specialisation and, eventually, the different outputs they produce. Not only do different outputs, and different combination of outputs, imply different technologies, but this differentiation is itself the consequence of the above-mentioned unobserved heterogeneity implied by external conditions (weather, land fertility, exposition to natural hazards, location, etc.). Farms typically are multi-output production units, and within a farm sample across a large geographical scale (like a country), hundreds of different agricultural products can be observed. Unfortunately, the stochastic frontier approaches proposed so far adopt technological specifications either that are single output (Greene, 2005; Njuki, Bravo-Ureta and O'Donnell, 2019; Lachaud, Bravo-Ureta and Ludena, 2017) or that imply strong output aggregation (e.g. O'Donnell, 2016).

In the end, the advantages of the stochastic frontier approach may be substantially limited, if not offset, by the disadvantages listed above. There is no reason to consider the econometric and the index number approach as competitors and no superiority of one approach over the other should be assumed (Hulten, 2001; OECD, 2001). The most appropriate approach depends on the characteristics of the panel data set and on the main research question (Antle and Capalbo, 1989: 62–63). Here, a farm panel data set with large cross-farm heterogeneity is used. Moreover, the research interest is not in measuring the total economic effects of immigrant workers in agriculture (which would require a full representation of the technology), but is exclusively in whether and how the presence and composition of immigrant workforce affect the

8 This is the case of the use of random coefficients estimation and of semi- or non-parametric approaches (Greene, 2005; Lachaud, Bravo-Ureta and Ludena, 2017; Njuki, Bravo-Ureta and O'Donnell, 2019).



farm-level TFP productivity in order to empirically verify the diversity hypothesis.<sup>9</sup> The index numbers approach allows this without trying to recover the underlying technology under a large and unknown farm-level heterogeneity.

The methodological approach thus consists of two stages. Firstly, the left-hand-side terms of equation (3) are measured as a TFP index through index number techniques. Secondly, this TFP is regressed on the right-hand-side terms, thus on some measure of labour diversity. This two-stage approach is frequent in the empirical literature whenever the interest is not in the production technology, but in assessing the determinants of TFP growth and differentials (Chambers, 1988: 232; O'Donnell, 2016: 328–329).

The main advantage of this two-stage approach is with allowing any farm to have its own specific technology, and in fully considering its heterogeneity. The well-known endogeneity issue implied by production function estimation (Griliches and Mairesse, 1995) remains a concern, but it is here limited to the second estimation stage and to term  $D$ , and not to  $K$  and  $L$ . In fact, on the right hand side of equation (3), the first term ( $\lambda$ ) behaves as a exogenous term, while the second term ( $d_{it}$ ) is clearly endogenous as it is associated with the labour factor share and, ultimately, is itself a consequence of the input use decisions that also affect the left-hand side of equation (3), i.e. the TFP (Hulten, 1992: 968).

Following this empirical strategy, the stochastic nature of productivity is only apparently lost in the TFP index measurement. In fact, the stochastic noise (i.e. errors and deviations as well as unobserved heterogeneity) is explicitly included in the second stage of the analysis where all the possible determinants of the observed TFP, the stochastic terms included, are made explicit (see Section 3.4).

### 3.3. Expressing and measuring productive diversity

To follow this empirical strategy, equation (3) must be reformulated in discrete-time, rather than continuous-time, variations. This reformulation is derived in Annex 1 in supplementary data where the productive diversity is expressed as  $\bar{D}_{it} = \ln \frac{D_{it}}{D_{it-1}}$ , i.e. as an index with respect to a reference observation. The productive diversity  $\bar{D}_{it}$  cannot be measured directly but is expressed as a composite indicator combining observable features of the farm workforce. Therefore, empirically, the construction of this indicator depends on the available information about these characteristics. Considering the recent literature, a measure of immigrant diversity should describe at least two elementary characteristics: (i) it must capture the degree to which the foreign born

9 It remains true that, in principle, a deeper understanding, and possibly a more correct estimation, of the role of immigrant workers on the farm-level TFP performance could be obtained by specifying and estimating a stochastic production frontier that makes all the substitution/complementary relationships among inputs and the stochastic noise explicit. Such a research effort, as discussed, still encounters several empirical issues but could represent a substantial step forward in this topic. In this respect, the conservative methodological approach here adopted aims to provide a preliminary empirical support to the diversity hypothesis on which future and more sophisticated studies could contribute with more comprehensive and accurate estimations.



are present and (ii) it should describe the range of source countries from which those individuals originate (Kemeny, 2017: 169).

Here, the composite indicator of farm-level workforce diversity is built on these two elementary variables: the share of immigrants (both permanent and seasonal) in the total farm labour force in annual working units (AWU) ( $S_{it}$ , with  $0 \leq S_{it} \leq 1$ ) and the ethnic heterogeneity within this immigrant labour force. Consistently with the literature on this topic (Easterly and Levine, 1997; Alesina *et al.*, 2003; Posner, 2004), the ethnic heterogeneity is represented by the ethno-linguistic fractionalisation (ELF) index reflecting the likelihood that two people randomly chosen within a population come from different ethnic groups. It is calculated using a simple Herfindahl concentration index,

$ELF_{it} = 1 - \sum_{h=1}^m s_{it,h}^2$ , where  $s_{it,h}^2$  is the quadratic share of the generic  $h$ -th ethnic group ( $h = 1, \dots, m$ ) observed in the  $i$ -th farm at time  $t$ . Following Alesina, Harnoss and Rapoport (2018), ethnic group is here country of origin.  $ELF_{it}$  is  $0 \leq ELF_{it} \leq 1, \forall i, t$  and it is small ( $\rightarrow 0$ ) when immigrants are homogenous from an ethnic point of view, while it is large ( $\rightarrow 1$ ) under ethnic heterogeneity.

Following the discussion above (see Section 2), the objective is to define a composite indicator where both aspects contribute to diversity in order to make the diversity hypothesis emerge but, at the same time, avoiding any constraint on estimated coefficients and facilitating their interpretability. More in detail, the diversity composite indicator here adopted is based on the normalisation of the two elementary variables, is parsimonious in parameters, allows compensability, but does not impose pure additivity between the two variables (OECD-JRC, 2008).

Normalisation is needed to render the variables comparable, and it is here granted by the fact that both  $S_{it}$  and  $ELF_{it}$  range between 0 and 1. Parsimony and compensability can be obtained by a linear aggregation of the two variables with positive weights adding up to one (unity sum property). These weights imply compensability, that is, a trade-off between the elementary variables. So, the estimation of the unknown weights can be interpreted as the relevance of the two variables on productive diversity. To avoid pure additivity, the linear aggregation can be augmented with an unconstrained (in sign and magnitude) interaction term, such that both sub- and super-additivity are admitted. The interaction term introduces non-perfect compensability (that is, sub- or super-additivity) and eventually also allows for a negative effect of both variables on diversity and then, on TFP (see below). In fact, the introduction of the interaction term makes the adopted aggregation a mix of linear and geometric aggregation, thus implies compensability but allows for non-constant compensability (OECD-JRC, 2008).

$\bar{D}_{it}$  is thus defined as follows:

$$\bar{D}_{it} = wS_{it} + (1 - w)ELF_{it} + \gamma S_{it}ELF_{it} \quad (4)$$

where  $w$  is an unknown weight with  $0 \leq w \leq 1$  and  $\gamma$  is a further unknown and unconstrained parameter.

Defined as in equation (4),  $\bar{D}_{it}$  is affected by both  $S_{it}$  and  $ELF_{it}$  but the respective effects are different and not constant as they depend on the weights and on the magnitude of the other variable. It is:  $\partial \bar{D}_{it} / \partial S_{it} = w + \gamma ELF_{it}$  and  $\partial \bar{D}_{it} / \partial ELF_{it} = (1 - w) + \gamma S_{it}$ . Sign and size of  $\gamma$  thus establishes whether the combination of a high share of immigrants and high ethnic fractionalisation reinforces the productive diversity. A negative-enough  $\gamma$  could imply that an increase in fractionalisation reduces productive diversity if the share of immigrants is large enough.

These properties of the composite indicator  $\bar{D}_{it}$  also imply some desirable properties in the way diversity affects TFP. These properties emerge clearly by inserting (4) into equation (3) once reformulated in discrete time (see Annex 1 in supplementary data). We obtain:

$$\ln TFP_{it} = \lambda + w\theta_{Lit}S_{it} + (1 - w)\theta_{Lit}ELF_{it} + \gamma\theta_{Lit}S_{it}ELF_{it} \quad (5)$$

It follows that  $\partial \ln TFP_{it} / \partial S_{it} = w\theta_{Lit} + \gamma\theta_{Lit}ELF_{it}$  and  $\partial \ln TFP_{it} / \partial ELF_{it} = (1 - w)\theta_{Lit} + \gamma\theta_{Lit}S_{it}$  indicating that both  $S_{it}$  and  $ELF_{it}$ , via a positive impact on productive diversity ( $\bar{D}_{it}$ ), positively affect the farm-level productivity unless their interaction is negative and large enough to offset this effect and reduce the productive diversity. Hence, the total differential of  $\ln TFP$  with respect to the variations of the two components of productive diversity,  $S$  and  $ELF$ , is:<sup>10</sup>

$$\begin{aligned} d \ln TFP_{it} &= w\theta_{Lit} + (1 - w)\theta_{Lit} + \gamma\theta_{Lit}(S_{it} + ELF_{it}) \\ &= \theta_{Lit}[1 + \gamma(S_{it} + ELF_{it})] \end{aligned} \quad (6)$$

Equation (6) reveals interesting properties of the composite indicator  $\bar{D}$  on  $TFP$  and, above all, leads to an empirically tractable formalisation of the diversity hypothesis. Firstly, the impact on TFP is null when  $\bar{D}_{it} = 0$ , i.e. when there are no immigrants within the farm labour force. Secondly, the impact on TFP may reach the maximum and possibly be greater than one at  $(1 + \gamma)$  with  $\gamma > 0$ , i.e. when all farm workers are immigrants and are all of different ethnic origin. Thirdly, due to non-perfect compensability between  $S$  and  $ELF$ , the productivity impact of their simultaneous and concordant variations does not depend on weight  $w$  but depends on their interaction.

Fourthly, and importantly here, equation (6) can be either positive or negative depending on  $\gamma$ ,  $S_{it}$  and  $ELF_{it}$ , and equation (5) allows for a non-monotonic relationship between diversity and productivity, thus admitting (but not imposing) the diversity hypothesis. The increase of  $S$  and/or  $ELF$  initially increases the productive diversity,  $\bar{D}$ , thus  $TFP$ . Then, it reaches a level of  $S$  and  $ELF$  such that a further increase generates a reduction of  $\bar{D}$ , thus that of  $TFP$ .

10 Since equation (5) is specified in the logarithm of TFP, TFP responds non-monotonically with respect to  $\bar{D}_{it}$  (the combination of  $S_{it}$  and  $ELF_{it}$ ). TFP is also non-linear with respect to both  $S_{it}$  and  $ELF_{it}$ . See also the results section.

Accordingly:

$$\begin{cases} d\ln TFP_{it} > 0 \text{ when } \gamma > 0 \text{ or } \gamma < 0 \text{ but } \gamma(S_{it} + ELF_{it}) > -1 \\ d\ln TFP_{it} < 0 \text{ when } \gamma < 0 \text{ and } \gamma(S_{it} + ELF_{it}) < -1 \end{cases} \quad (7)$$

Therefore, the necessary though not sufficient condition to have a non-monotonic relationship between  $TFP_{it}$  and  $(S_{it} + ELF_{it})$  is  $\gamma < 0$ . In fact, with  $\gamma > 0$ , only a positive relationship between  $TFP_{it}$  and  $(S_{it} + ELF_{it})$  emerges, while with  $\gamma < 0$ , both positive and negative signs of the relationship may occur. Given  $\gamma < 0$ , the relationship will be positive for small-enough diversity (i.e.  $S_{it} + ELF_{it}$ ) and negative otherwise. Consequently, according to the diversity hypothesis discussed in Section 2 and to the derivation presented,  $S_{it}$  and  $ELF_{it}$  are expected to have a positive effect on productive diversity and, thus, on farm TFP, up to the point where  $\gamma(S_{it} + ELF_{it}) > -1$ . However, when diversity becomes too high, i.e. when  $\gamma(S_{it} + ELF_{it}) < -1$ , it becomes unproductive and their combination eventually produces a negative effect on TFP.

### 3.4. Making TFP stochastic

Within the farm sector, large heterogeneity is usually observed across units in terms of TFP (Baldoni, Codaroni and Esposti, 2018). To allow for both this heterogeneity and the stochastic nature of the farm-level productivity equation (6) can be augmented. Firstly, the Hicks-neutral technological change ( $\lambda$ ) can be replaced as follows:

$$\ln TFP_{it} = \mu_0 + \mu_t + \mu_i + \mu_{it} + w\theta_{Lit}S_{it} + (1-w)\theta_{Lit}ELF_{it} + \gamma\theta_{Lit}S_{it}ELF_{it} \quad (8)$$

where  $\lambda = \mu_0 + \mu_t + \mu_i + \mu_{it}$ .

Equation (8) makes explicit all the possible sources of heterogeneity of this exogenous component of farm productivity within the panel:  $\mu_0$  represents the mean level across farms and over time;  $\mu_t$  represents the  $t$ -th time specific level common to all farms;  $\mu_i$  represents the  $i$ -th farm time-invariant specific level and finally,  $\mu_{it}$  represents the  $i$ -th farm time-variant specific level. While the exogenous productivity component is unobserved, not all its farm-level determinants are. Eberhardt and Helmers (2010) propose the following general specification to explicitly distinguish between observed and unobserved productivity determinants:

$$\ln TFP_{it} = \mu_0 + \mu_t + \mu_i + \mu_{it} + w\theta_{Lit}S_{it} + (1-w)\theta_{Lit}ELF_{it} + \gamma\theta_{Lit}S_{it}ELF_{it} + \mathbf{Z}_i\mathbf{\Gamma} + \mathbf{X}_{it}\mathbf{\Pi} \quad (9)$$

where  $\mathbf{Z}_i$  and  $\mathbf{X}_{it}$  are  $(1 \times k)$  and  $(1 \times h)$  vectors of time invariant and time-variant observable productivity determinants, respectively, and  $\mathbf{\Gamma}$  and  $\mathbf{\Pi}$  are the correspondent  $(k \times 1)$  and  $(h \times 1)$  vectors of unknown parameters to be estimated.

As Griliches and Mairesse (1995) point out,  $\mu_{it}$  is unobserved by the econometrician but is known by the producer. So, from the econometrician's

perspective, it simply represents a component of the error term while, for the farmer, it is available information (expressing some time-variant firm/farm characteristic or condition) that affects input use decisions. Therefore, inputs are partially determined by the unobserved time-variable characteristics contained in  $\mu_{it}$ , so the usual exogeneity assumptions are unlikely to hold. [Blundell and Bond \(2000\)](#) propose a theoretically and econometrically consistent solution to this endogeneity problem by specifying the dynamic stochastic process generating  $\mu_{it}$ ,  $\forall i, t$ :

$$\mu_{it} = \rho\mu_{it-1} + \varepsilon_{it} \quad (10)$$

where  $\varepsilon_{it}$  is an i.i.d.  $\sim N(0, \sigma^2)$  error term<sup>11</sup> representing deviations from the productivity mean due to measurement errors, unexpected delays or other external circumstances ([Van Beveren, 2010](#)). The introduction of the AR(1) term aims to capture the impact of past productivity shocks on current input decisions, so it internalises that producers might react with delay to changes in productivity. This delay occurs because input decisions are subject to hiring/investment costs generating rigidities in the response. Within a neoclassic dynamic production function framework, [Bond and Söderbom \(2005\)](#) derive this autoregressive structure of term  $\mu_{it}$  as the outcome of the intertemporal optimisation problem of the producers under stochastic adjustment costs associated to any modification of input levels.

From equation (9) follows that  $\mu_{it-1} = \ln TFP_{it-1} - \mu_0 - \mu_i - \mu_{t-1} - w\theta_{Lit-1}S_{it-1} - (1-w)\theta_{Lit-1}ELF_{it-1} - \gamma\theta_{Lit-1}S_{it-1}ELF_{it-1} - \mathbf{Z}_i\mathbf{\Gamma} - \mathbf{X}_{it-1}\mathbf{\Pi}$ , by replacing in equation (10) and rearranging we obtain:

$$\begin{aligned} \ln TFP_{it} &= \bar{\mu}_0 + \bar{\mu}_t + \bar{\mu}_i + \rho \ln TFP_{it-1} + w\theta_{Lit}S_{it} + (1-w)\theta_{Lit}ELF_{it} \\ &+ \gamma\theta_{Lit}S_{it}ELF_{it} + \bar{w}\theta_{Lit-1}S_{it-1} + \bar{\bar{w}}\theta_{Lit-1}ELF_{it-1} \\ &+ \bar{\gamma}\theta_{Lit-1}S_{it-1}ELF_{it-1} + \mathbf{Z}_i\bar{\mathbf{\Gamma}} + \mathbf{X}_{it}\mathbf{\Pi} + \mathbf{X}_{it-1}\bar{\mathbf{\Pi}} + \varepsilon_{it} \end{aligned} \quad (11)$$

where  $\bar{\mu}_0 = (1-\rho)\mu_0$ ,  $\bar{\mu}_t = \mu_t - \rho\mu_{t-1}$ ,  $\bar{\mu}_i = (1-\rho)\mu_i$ ,  $\bar{w} = -\rho w$ ,  $\bar{\bar{w}} = -\rho(1-w)$ ,  $\bar{\gamma} = -\rho\gamma$ ,  $\bar{\mathbf{\Gamma}} = (1-\rho)\mathbf{\Gamma}$ ,  $\bar{\mathbf{\Pi}} = -\rho\mathbf{\Pi}$ .

Restrictions on parameters associated with terms  $\mathbf{X}_{it}$  and  $\mathbf{X}_{it-1}$  and terms  $(\theta_{Lit-1}ELF_{it-1} - \theta_{Lit}S_{it-1})$  and  $\theta_{Lit-1}S_{it-1}ELF_{it-1}$  cannot be imposed *ex ante* and have to be statistically tested post estimation. To impose parameter

11 The normality assumption on this disturbance term implies that the dependent variable itself is assumed to be normally distributed. This assumption is clearly violated for the TFP level due to the strong asymmetry of productivity within the sample as it ranges in the  $[0, \infty)$  interval (cases with zero TFP are evidently lacking; see Table A1 in supplementary data for more descriptive evidence). For this same reason, however, its log-normal distribution over the  $[0, \infty)$  support can be largely accepted. Since model equation (11) is specified in terms of  $\ln TFP$ , normality seems a reasonable assumption here.

restrictions, equation (11) can be rewritten as follows:

$$\begin{aligned} \ln TFP_{it} = & \bar{\mu}_0 + \bar{\mu}_t + \bar{\mu}_i + \rho (\ln TFP_{it-1} - \theta_{Lit-1} ELF_{it-1}) \\ & + w(\theta_{Lit} S_{it} - \theta_{Lit} ELF_{it}) + \theta_{Lit} ELF_{it} + \gamma \theta_{Lit} S_{it} ELF_{it} \\ & + \rho w(\theta_{Lit-1} ELF_{it-1} - \theta_{Lit-1} S_{it-1}) - \rho \gamma \theta_{Lit-1} S_{it-1} ELF_{it-1} \\ & + \mathbf{Z}_i \bar{\Gamma} + \mathbf{X}_{it} \Pi + \mathbf{X}_{it-1} \bar{\Pi} + \varepsilon_{it} \end{aligned} \quad (12)$$

### 3.5. The empirical model specification and estimation

Equations (11) and (12) are dynamic (AR(1)) panel models whose estimation, once farm TFP levels are computed, allows to directly assess whether the farm's share of immigrant workforce ( $S_{it}$ ) and its composition ( $ELF_{it}$ ) influence the farm's productivity. The TFP level also depends on other observed farm characteristics, either time-invariant ( $\mathbf{Z}_i$ ) or time-variant ( $\bar{\mathbf{X}}_{it}$ ). Here,  $\mathbf{Z}_i = (R_{i,r}, Alt_i, Age_i)$  where  $R_r$  are regional dummies (where  $r$  indicates the region);  $Age$  is the age of the farm holder (average over the period);  $Alt$  is the average altitude of the farm land (average over the period);<sup>12</sup>  $\bar{\mathbf{X}}_{it} = (ES_{it}, P_{it,f})$  where  $ES_{it}$  is the farm's economic size (ES) expressed by the gross production value;  $P_{it,f}$  is the farm specialisation expressed by the  $f$ -th product's share on farm's value of production, with  $f$  = dairy, cereals, livestock, fruits, vegetables, wine and others.

The dynamic panel specification was estimated using both the Difference Generalized Method of Moments (GMM-DIF) and the System Generalized Method of Moments (GMM-SYS) estimator (Arellano, 2003). The two-step GMM estimation was chosen for its asymptotic properties, and Windmeijer's correction for the standard error small sample bias was adopted (Arellano, 2003). The GMM-SYS estimation is expected to overcome the so-called weak instruments problem, especially when  $\rho \rightarrow 1$ , and, therefore, to provide better results. Its superiority in the finite sample case, however, is questionable, and the risk of incurring overidentifying restrictions is higher.<sup>13</sup> The unrestricted model (11) is firstly estimated, then restrictions are imposed, and the respective specification is estimated (12). The comparison of the restricted and unrestricted model allows to better appreciate how the theoretical restrictions affect the results and to what extent they may be considered statistically acceptable.

12 The altitude may be a time-varying variable as the average altitude of the plots of any farm may vary because their land endowment may change over time. Nonetheless, these variations are minimal so here altitude is considered a time-invariant value taken as the average altitude over the period 2008–2015.

13 Due to first-differentiation, time-invariant regressors ( $R_{i,r}$ ,  $Alt_i$ ,  $Age_i$ ) are cancelled out in the GMM-DIF estimation stage. In principle, the use of the model expressed in the levels instrumented by the lagged first differences allows the GMM-SYS approach to also estimate the coefficients of the time-invariant variables. However, in the case of a fixed effects specification, as in the present case, this remains unfeasible (Roodman, 2009). Very recent studies propose two-step GMM estimation approaches that also estimate the coefficients of the time-invariant regressors (Kripfganz and Schwarz, 2019). However, the small-sample performance of these new approaches with respect to the time-variant regressors, of interest here, is still under investigation. The use of these recent two-step procedures may represent an interesting opportunity for future research.

All these estimation approaches may encounter the possible endogeneity of some of the regressors. It cannot be excluded that productivity depends on the producer's decisions expressed by these regressors that can themselves be dependent on productivity, as they are observed by the farmer. As already discussed, this can be the case for  $S_{it}$  and  $ELF_{it}$  which may express the farmer's choice to change quality and composition of the labour force as a consequence of the TFP performance. Analogously, the farm size,  $ES_{it}$ , may express the farmer's choices depending on current and past productivity levels. Moreover, as size expresses the possible returns to scale at the farm level, this variable may be reflected in the index number TFP calculation. Consequently, for  $S_{it}$ ,  $ELF_{it}$  and  $ES_{it}$ , the assumption of exogeneity can be questionable. Therefore, all the above-mentioned estimations are performed by instrumenting these variables with the respective suitable lagged values.

## 4. Data

### 4.1. The Italian case study

From the late 1970s onwards, immigrant workers have become an important part of the Italian agricultural workforce (Kasimis, 2006). In 2017, this share was 17 per cent according to official statistics (which excludes irregular and illegal workers) (ISTAT, 2018). The most recent agricultural census (2010) gives the share as 25 per cent for farms with salaried workers (Coderoni *et al.*, 2018).

Italian agriculture represents an ideal case for investigating the relationship between immigrant workers and productivity. This workforce consists of predominantly unskilled workers (Gertel and Sippel, 2014) and the actual contribution of the immigrant workforce to the Italian economy is controversial (Bettin, Lo Turco and Maggioni, 2014; Bratti and Conti, 2018). Besides, the Italian agricultural sector is highly heterogeneous in terms of farming systems, production specialisations, size and location, productivity levels and presence and composition of the immigrant workforce (Baldoni, Coderoni and Esposti, 2017).

Two practical issues complicate the empirical assessment of the role of the immigrant workforce in Italian agriculture. The first concerns the coverage of the official statistics. The second deals with the appropriate level of investigation (level of aggregation of farm data). It can be argued that using the official statistics data might provide an inaccurate representation of the phenomenon because these barely capture the presence of irregular workers. Irregular workers are often occupied in seasonal activities (MAC, 2013), and the use of official statistics could bias the measure of the presence of immigrants for these seasonal activities. The relevant empirical issue here is whether the presence of foreign workers could be systematically underestimated and geographically biased. According to the last ISTAT estimates, the number of irregulars employed in agriculture in the north (43 per cent of total irregulars) and in the south (45 per cent) is very similar, representing also similar shares of total labour units (26 per cent in the north and 24 per cent in the south) (ISTAT,

2018). Therefore, even though there is a large agreement on the fact that the immigrant workforce in Italian agriculture can be significantly underestimated by official data, the share of irregular workforce of total workforce seems to be quite equally distributed across regions making these data still reliable to describe key features.

The presence of immigrant workforce in Italian agriculture is highly differentiated across farm typologies and mostly concentrated in professional and market-oriented farms (Coderoni *et al.*, 2018). Aggregated data at the regional level may only capture part of this heterogeneity because of the large presence of the non-professional farms. It seems appropriate to investigate the impact of immigrant workers on Italian agricultural productivity by considering farm-level data allowing the above-mentioned heterogeneity to emerge and then to limit the analysis to professional and market-oriented activities. For this reason, the present analysis uses the Italian Farm Accountancy Data Network (FADN) data set that is representative of farms with an ES over an established threshold (EUR 4,800 of yearly standard output-SO-up to 2013; EUR 8,000 afterwards). Annex 2 in the supplementary data provides a description of the FADN survey detailing the information collected with respect to immigrant workforce.

## 4.2. The data set

The 2008–2015 Italian FADN data set consists of an unbalanced panel of farms ranging from 10,693 farms (2011) to 7,649 farms (2015), with a total amount of 79,544 observations over the whole period. From this data set, a balanced panel of 1,935 farms was extracted, thus a total sample of 15,480 observations. A major advantage of using a balanced panel is that it eliminates the risk that the observed farm-level productivity and presence of immigrant workforce is influenced by the change in the composition of the sample and, thus, by the different productivity and presence of immigrants of the farms entering/exiting the sample, with respect to the existing ones. For our application, the balanced panel did not lead to a significant loss of information compared with using the unbalanced sample. Firstly, given the model specification, only farms with at least three consecutive observations can be used. This would reduce the unbalanced panel to 57,045 observations. Secondly, data do not differ much between the two samples, except for the presence of few larger outliers in the unbalanced sample. Table A1 in supplementary data compares the model variables' descriptive statistics between the two samples. The values of the variables of interest (share and ethnic heterogeneity of immigrants) together with some structural variables (output elasticity of labour and farmer's age) are similar as indicated by the *t*-test on the samples' means, while some small difference emerges for performance variables like *TFP* and *ES*. In these cases, the unbalanced panel shows distributions with a longer right tail because of the presence of some large and very productive farms particularly in the dairy, cereal and fruit production. These farms exit and/or enter the FADN sample during the period under investigation due to the FADN sampling scheme. Because of this



**Table 1.** Working units and shares of total workforce and immigrant workforce in the 2008–2015 Italian FADN balanced panel.

|                                        | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | %variation<br>2008/2015 |
|----------------------------------------|------|------|------|------|------|------|------|------|-------------------------|
| Employees (AWU)                        | 752  | 732  | 725  | 756  | 735  | 775  | 769  | 765  | 2                       |
| Immigrants (AWU)                       | 135  | 157  | 173  | 205  | 182  | 241  | 258  | 260  | 93                      |
| Immigrants seasonal (AWU)              | 100  | 112  | 130  | 162  | 130  | 185  | 196  | 192  | 91                      |
| Total seasonal (AWU)                   | 570  | 532  | 531  | 533  | 503  | 541  | 512  | 513  | –10                     |
| Immigrants/employees (%)               | 18   | 21   | 24   | 27   | 25   | 31   | 33   | 34   | 90                      |
| Immigrants seasonal/total seasonal (%) | 18   | 21   | 24   | 30   | 26   | 34   | 38   | 37   | 113                     |
| Immigrants seasonal/immigrants (%)     | 74   | 71   | 75   | 79   | 71   | 77   | 76   | 74   | –1                      |

peculiarity, it seems preferable here to estimate the model on the balanced panel to obtain more robust evidence.<sup>14</sup>

Table 1 reports the AWU and shares of workers within the farm panel by year, focusing on immigrants, and Table A3 in supplementary data provides their allocation across farm specialisation and size classes for 2008 and 2015. In the panel, the immigrant workforce constituted 26 per cent of the annual total salaried workforce over the 2008–2015 period on average. This share is in line with the latest (2010) Italian Agricultural Census where the immigrant workforce represented 25 per cent of the total agricultural salaried workforce (Coderoni *et al.*, 2018).

Table 1 further shows that the number of immigrants' AWU increased by about 93 per cent from 2008 to 2015 and the respective share on total salaried occupation by 90 per cent.<sup>15</sup> Seasonal workers, as expected, represent on

14 Also, the structure of the two samples are very similar in terms of shares of different farm type and size as shown in Table A2 in supplementary data. Model estimates obtained on the unbalanced panel are fully concordant with those obtained on the balanced panel but show lower statistical quality. These estimates are available upon request.

15 Since 2016, Italy has experienced major waves of sea arrivals, as shown by data of the United Nations High Commissioner for Refugees (<https://data2.unhcr.org/en/situations/mediterranean/location/5205>. Accessed on 29 February 2020). However, these arrivals are not directly linked to the participation of immigrants to Italian agricultural workforce. In fact, data on the number of immigrant agricultural workers in Italy, provided by the National Social Security Institute (INPS), show that their presence over the period 2008–2017 remained almost unchanged, with a slight

average 75 per cent of total immigrant AWU. This percentage is almost stable over 2008–2015 and, consequently, immigrants' share in the total seasonal workforce has more than doubled during this period.

With 10 per cent, the share of farms employing immigrant workers is still quite small. Looking at the distribution of these immigrant workers across farm specialisations and size classes, the large heterogeneity within the sample clearly emerges. The higher shares of immigrants in 2015 are found in fruit and wine production and the dairy sector followed by arable crops, while they are much lower in granivores, mixed and olives specialisations. These shares have not changed much since 2008, except for wine specialists. This heterogeneous distribution across farms leads to the geographical distribution of the immigrant workforce (see also Baldoni, Coderoni and Esposti, 2017: 268).

As for size classes, farms are grouped into three categories: small farms (with a SO less than EUR 25,000), medium farms (with a SO between EUR 25,000 and 100,000) and large farms (with a SO higher than EUR 100,000). As highlighted in previous work (Malchow-Møller *et al.*, 2013), medium and large farms employ relatively more immigrant workers.

The immigrant workers in the panel are mainly from EU countries. In 2015, 37 per cent of the total immigrant workforce in Italian agriculture was from a single EU country, Romania. The immigrant workforce from Slovakia and Czech Republic exhibited fast growth rates over the period, though these shares remained small. Asia and Africa were the second and third most important areas of origin of immigrants, respectively.

Finally, the panel data show that in 2008 the ELF index was highest for olive, dairy and fruit specialists, while in 2015 the highest values were observed in fruit, dairy, livestock and horticulture. Over the period, the index increased by about 40 per cent.

#### 4.3. The farm-level TFP measure

The TFP levels to be used in the estimation of equations (11)–(12) were computed for any farm  $i$  in time period  $t$ , relative to farm  $j$  in time period  $s$ , using a multilateral index number approach. This relative TFP measure is obtained as ratio of an output quantity index ( $QI_{it,js}^O$ ) to an input quantity index ( $QI_{it,js}^I$ ) (Coelli *et al.*, 2005):

$$TFP_{it,js} = \frac{QI_{it,js}^O}{QI_{it,js}^I}, \forall i, j \in N, i \neq j \text{ and } \forall t, s \in T, t \neq s. \quad (13)$$

A major complication arises when using conventional superlative indexes within panel comparisons, since measurements might not satisfy the transitivity property (Rao, O'Donnell and Ball, 2002), i.e. a direct comparison between two farms might not be equal to the indirect comparisons of the two through a

2 per cent increase (De Leo and Vanino, 2019). This is mainly because, in Italy, the asylum seekers who wait for the resolution of an application for international protection for 2/3 years, are meanwhile hosted in dedicated centres, but very rarely become agricultural workers (at least regular workers, of interest here).

third one. This property is extremely important because it ensures the internal consistency and the uniqueness of results (Hill, 2004). To address this transitivity issue, the minimum spanning tree (MST) method proposed by Hill (1999) is used. This method is based on chaining a sequence of bilateral comparisons. However, within a panel data set, there is no natural/univocal ordering for the chaining procedure. We follow Hill (1999, 2004) and selected the most reliable among all possible bilateral comparisons based on the Paasche–Laspeyres spread (PLS). The Paasche and Laspeyres index number formulas ‘provide bounds for the true index that is defined using economic theory’ (Coelli *et al.*, 2005), and hence, a small PLS is desirable because it implies that index number aggregation is less sensitive to the choice of formula (Hill, 1999, 2004). The PLS between two farms in any year is a distance function that is zero in the case their vectors of output or input quantities are proportional. The spread will be small whenever the production structure of two farms is similar, i.e. where farms supply/demand similar products/production factors. After the MST had been identified, the Fisher index was used to chain the bilateral comparisons and derive transitive output and input quantity indexes.

The output index is created using the quantity and value information on 298 crop and livestock products. The Italian FADN data set contains information on the respective quantity produced and total values. Implicit prices are obtained dividing total values by the quantity produced. The input index aggregates the following factors of production: labour, fertilisers, pesticides, external services, water, energy, seeds, feeding stuff, capital, land, reuses and other costs. With respect to labour, the Italian FADN provides data on the hourly opportunity cost for labour. Total salary for family labour is estimated by multiplying their number of hours worked by this opportunity cost.

For capital, an index of capital services was created adapting the Organisation for Economic Co-operation and Development (OECD) capital measurement methodology (OECD, 2009) for farm-level data. Assets include machinery, buildings, plantations and livestock. Information on their value and expected useful life are contained in the Italian FADN data set and were used to estimate their productive stocks, assuming a hyperbolic efficiency-loss function with a shape parameter that varies by asset type (Rizzi and Pierani, 2006), and to estimate user costs. User costs were computed as a function of an exogenous rate of return, depreciation and expected asset price revaluation. The average annual yield of Italian government bonds with 10-year maturity between 2002 and 2015 was used as the exogenous rate of return. Using the Fisher formula, productive stock by asset was aggregated to obtain an index of capital services using user costs as weights.

Table A1 in supplementary data reports the descriptive statistics of this farm-level TFP calculation in both the unbalanced and balanced FADN Italian panel, while Annex 3 reports detailed data description and additional information.

**Table 2.** Dynamic (GMM-DIF and GMM-SYS) unrestricted model estimations (standard errors in parenthesis).

|                                        | GMM-DIF          | GMM-SYS          |
|----------------------------------------|------------------|------------------|
| $\ln TFP_{it-1}$                       | 0.110 (0.022)*** | 0.105 (0.023)*** |
| $\theta_{Lit}S_{it}$                   | -1.650 (1.593)*  | -2.160 (1.147)** |
| $\theta_{Lit-1}S_{it-1}$               | 1.287 (3.192)    | 0.204 (0.891)    |
| $\theta_{Lit}ELF_{it}$                 | -0.715 (0.413)*  | -0.563 (0.371)   |
| $\theta_{Lit-1}ELF_{it-1}$             | -0.079 (0.460)   | 0.150 (0.318)    |
| $\theta_{Lit}S_{it}ELF_{it}$           | 1.614 (1.567)    | 1.932 (1.170)*   |
| $\theta_{Lit-1}S_{it-1}ELF_{it-1}$     | -1.796 (2.540)   | -0.996 (1.422)   |
| $ES_{it}$                              | 0.594 (0.326)*   | 0.479 (0.290)*   |
| $ES_{it-1}$                            | 0.041 (0.072)    | 0.024 (0.060)    |
| $P_{it,Dairy}$                         | 1.410 (0.506)*** | 1.330 (0.470)*** |
| $P_{it,Cereals}$                       | 1.860 (0.702)*** | 1.690 (0.595)*** |
| $P_{it,Livestock}$                     | 1.510 (0.620)**  | 1.373 (0.546)**  |
| $P_{it,Fruits}$                        | 1.534 (0.808)*   | 1.458 (0.724)**  |
| $P_{it,Vegetables}$                    | 1.576 (0.459)*** | 1.536 (0.420)*** |
| $P_{it,Wine}$                          | 1.822 (0.694)*** | 1.700 (0.636)*** |
| $P_{it,Others}$                        | (omitted)        | (omitted)        |
| $P_{it-1,Dairy}$                       | -0.178 (0.494)   | -0.072 (0.488)   |
| $P_{it-1,Cereals}$                     | -0.177 (0.458)   | -0.074 (0.452)   |
| $P_{it-1,Livestock}$                   | -0.129 (0.461)   | -0.008 (0.458)   |
| $P_{it-1,Fruits}$                      | -0.656 (0.509)   | -0.438 (0.511)   |
| $P_{it-1,Vegetables}$                  | -0.301 (0.445)   | -0.163 (0.438)   |
| $P_{it-1,Wine}$                        | -0.290 (0.460)   | -0.145 (0.469)   |
| $P_{it-1,Others}$                      | (omitted)        | (omitted)        |
| Num. obs.                              | 7,740            | 13,545           |
| Sargan test ( $\chi^2$ )               | 14.63            | 24.88**          |
| Hansen test ( $\chi^2$ )               | 12.92            | 12.68            |
| Difference-in-Hansen test ( $\chi^2$ ) | -                | 3.96             |
| AR(1)                                  | -20.49***        | -20.40***        |
| AR(2)                                  | -0.98            | -1.08            |

\*, \*\* and \*\*\* represent statistically significant values at 1 per cent, 5 per cent and 10 per cent, respectively.

## 5. Results and discussion

### 5.1. Model estimates

The dynamic model specification is expected to take both heterogeneity and endogeneity properly into account as well as account for the dynamics inherent in farm-level productivity. Table 2 reports the results of the GMM-DIF and GMM-SYS estimation of the unrestricted dynamic model. Overall, results clearly suggest that the lagged TFP value matters. Sign and magnitude of this autocorrelation coefficient are consistent with what was obtained by [Esposti \(2000\)](#) for the Italian agriculture but using aggregate data. Other variables significantly affect the farm-level TFP.

Size positively influences productive performance, thus suggesting the presence of increasing returns to scale incorporated in the adopted TFP calculation. As expected, also production specialisation matters as all the respective variables are statistically significant. It follows implicitly that any investigation disregarding the specialisation heterogeneity may yield incorrect conclusions about farm-level TFP determinants. It also follows that the dynamic specification does not necessarily eradicate the effect of farm heterogeneity in this respect. The magnitude itself of these specialisation coefficients, while remarkable, requires careful interpretation. As these variables express the share of a specific production activity in the farm's value of production, a unit increase of one production activity necessarily implies a unit decrease of (some of) the other activities. Thus, what matters is the relative magnitude of these coefficients, and this suggests that TFP performance tends to be higher in wine and cereal production and lower in dairy and livestock. For all these variables, the estimated coefficients associated with the lagged values are not statistically significant, suggesting that the lagged TFP is able to capture all the autocorrelation within the data generation process.

Of main interest are the results for the presence of immigrant workers ( $S$ ) and of their composition ( $ELF$ ). For both variables, lagged values are not statistically significant, while current values are statistically significant for both  $S$  and  $ELF$  in the case of GMM-DIF estimation and for  $S$  and the interaction term in the GMM-SYS estimation. The magnitude of these coefficients must be carefully interpreted given the log-linear form of the estimated model. An estimated coefficient for  $S$  of 0.5 for instance implies that a 1 per cent increase in the share of immigrants on total workforce induces a 0.5 per cent increase in the value of the respective TFP index.

Sign and magnitude of the estimated coefficients, thus, do not facilitate the interpretation of these results in terms of impact of immigrant workforce share and composition on the farm-level TFP. The negative signs of  $S$  and  $ELF$  coefficients would contrast with any hypothesis of a positive impact of immigrant workforce on productivity. At the same time, however, the interaction between these two variables always shows a positive coefficient, whose magnitude could compensate the above-mentioned negative effect. Eventually, though difficult to interpret with respect to the theoretical justification of the diversity hypothesis (see Sections 2 and 3), these results seem to confirm the complex relationship between  $S$ ,  $ELF$  and  $TFP$ .

## 5.2. Assessing the diversity hypothesis

In order to better detect this complex relationship and to assess the diversity hypothesis, the theoretical structure from which this complexity and hypothesis emerge must be imposed. This is done by estimating the restricted version of equation (12). Table 3 reports the GMM-DIF and GMM-SYS estimates. In this estimation stage, parameter  $w$  is restricted to the interval  $0 < \hat{w} < 1$ .

The estimation results in Table 3 are mostly in line with those of the unrestricted model. This conclusion applied for  $ES$  and variables expressing production specialisation. Also, the role of the lagged TFP as a major driver of the current TFP is confirmed. On the basis of equation (12), the coefficient

**Table 3.** Dynamic (GMM-DIF and GMM-SYS) restricted model estimations (standard errors in parenthesis).

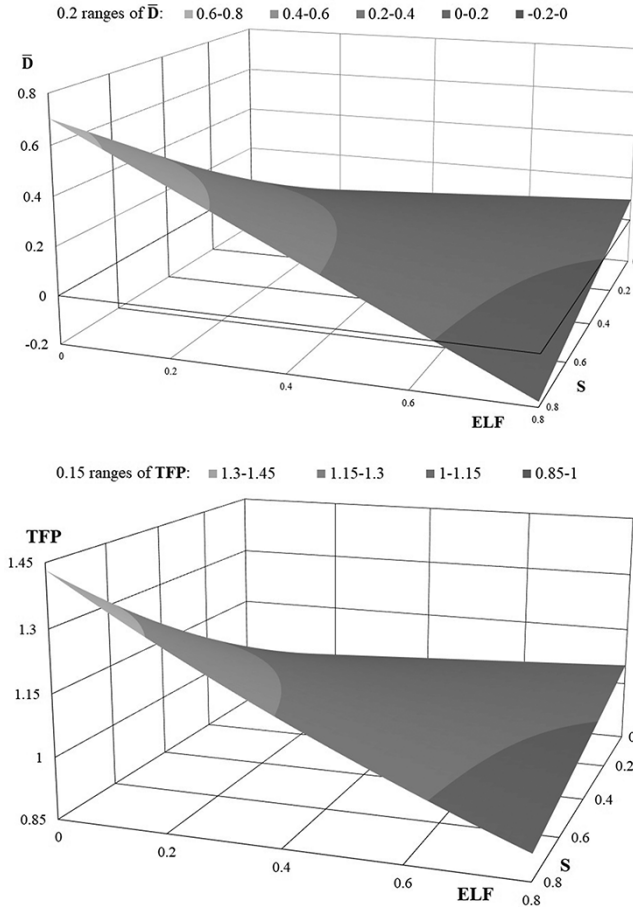
|                                                     | GMM-DIF          | GMM-SYS          |
|-----------------------------------------------------|------------------|------------------|
| $\ln TFP_{it-1} - \theta_{Lit-1} ELF_{it-1}$        | 0.130 (0.028)*** | 0.108 (0.023)*** |
| $\theta_{Lit} S_{it} - \theta_{Lit} ELF_{it}$       | 0.893 (0.187)*** | 0.878 (0.175)*** |
| $\theta_{Lit} S_{it} ELF_{it}$                      | -1.476 (0.703)** | -1.532 (0.642)** |
| $\theta_{Lit-1} ELF_{it-1} - \theta_{Lit} S_{it-1}$ | 0.043 (0.175)    | 0.011 (0.162)    |
| $\theta_{Lit-1} S_{it-1} ELF_{it-1}$                | -0.332 (1.256)   | -0.191 (0.575)   |
| $ES_{it}$                                           | 0.332 (0.440)    | 0.644 (0.375)*   |
| $ES_{it-1}$                                         | -0.017 (0.084)   | 0.037 (0.070)    |
| $P_{it,Dairy}$                                      | 1.219 (0.498)*** | 1.396 (0.519)*** |
| $P_{it,Cereals}$                                    | 1.443 (0.733)*   | 1.919 (0.711)*** |
| $P_{it,Livestock}$                                  | 1.169 (0.671)*   | 1.576 (0.637)**  |
| $P_{it,Fruits}$                                     | 1.061 (0.927)    | 1.765 (0.826)**  |
| $P_{it,Vegetables}$                                 | 1.398 (0.462)*** | 1.621 (0.469)*** |
| $P_{it,Wine}$                                       | 1.447 (0.785)*   | 1.975 (0.723)*** |
| $P_{it,Others}$                                     | (omitted)        | (omitted)        |
| $P_{it-1,Dairy}$                                    | -0.289 (0.490)   | -0.069 (0.486)   |
| $P_{it-1,Cereals}$                                  | -0.312 (0.456)   | -0.095 (0.447)   |
| $P_{it-1,Livestock}$                                | -0.212 (0.451)   | -0.060 (0.456)   |
| $P_{it-1,Fruits}$                                   | -0.753 (0.499)   | -0.497 (0.508)   |
| $P_{it-1,Vegetables}$                               | -0.379 (0.434)   | -0.219 (0.437)   |
| $P_{it-1,Wine}$                                     | -0.413 (0.460)   | -0.253 (0.468)   |
| $P_{it-1,Others}$                                   | (omitted)        | (omitted)        |
| Num. obs.                                           | 7,740            | 13,545           |
| Sargan test ( $\chi^2$ )                            | 8.65*            | 14.70            |
| Hansen test ( $\chi^2$ )                            | 7.75             | 12.94            |
| Difference-in-Hansen test ( $\chi^2$ )              | -                | 3.22             |
| AR(1)                                               | -18.40***        | -2040***         |
| AR(2)                                               | -0.56            | -1.03            |

\*, \*\* and \*\*\* represent statistically significant values at 1 per cent, 5 per cent and 10 per cent, respectively.

associated with the term  $(\ln TFP_{it-1} - \theta_{Lit-1} ELF_{it-1})$  expresses the autocorrelation term  $\rho$  whose estimate is fully consistent with that obtained in the unrestricted model.<sup>16</sup>

A main difference between the unrestricted and the restricted versions of the model emerges for the variables  $S$  and  $ELF$  for which, in fact, theoretical restrictions apply. The estimate of parameter  $w$  (associated with the term

16 According to the model theoretical derivation, the parameters associated to terms  $(\theta_{Lit-1} ELF_{it-1} - \theta_{Lit-1} S_{it-1})$  and  $(\theta_{Lit-1} S_{it-1} ELF_{it-1})$  should correspond to  $w$  and  $\gamma$ . In fact, by combining the estimates of  $\rho$ ,  $w$  and  $\gamma$ , we should obtain the following values with the respective standard error:  $w = 0.116$  (0.240) and  $\gamma = -0.192$  (0.804) for the GMM-DIF estimation;  $w = 0.095$  (0.218) and  $\gamma = -0.165$  (0.726) for the GMM-SYS estimation. In sign, magnitude and, statistical significance these combined parameters are consistent with the respective direct estimates ( $w = 0.043$  (0.175) and  $\gamma = -0.332$  (0.440) for the GMM-DIF estimation; ( $w = 0.011$  (0.162) and  $\gamma = -0.191$  (0.575) for the GMM-SYS estimation). Tests results are available upon request.



**Fig. 1.** Relation between the share of immigrant workforce ( $S$ ), its ethnic composition ( $ELF$ ) and the farm-level productive diversity ( $\bar{D}$ ) and productivity ( $TFP$ ).

*Notes:* The functional relationships are obtained according to equations (4) and (5), respectively, and the GMM-SYS model estimates (the sample average of the labour share is considered). The shades identify the combinations of  $ELF$  and  $S$  that imply values of  $\bar{D}$  and  $TFP$  falling in the respective range. The darkest areas identify a negative impact on  $\bar{D}$  and  $TFP$  ( $TFP = 1$  as starting value).

$(\theta_{Lit}S_{it} - \theta_{Lit}ELF_{it})$  is fully consistent with expectations (significantly positive but  $<1$ ) and would indicate that in the definition of productive diversity the share of the immigrant workforce assumes a much greater relevance than its composition. But it is the statistically significant and negative estimate of parameter  $\gamma$  (associated with the term  $\theta_{Lit}S_{it}ELF_{it}$ ) that allows to complete the picture of the influence of  $S$  and  $ELF$  on  $TFP$ . Farm-level  $TFP$  can respond either positively or negatively to a variation of the presence or composition of immigrants (i.e.  $S + ELF$ ), see equations (6)–(7). Consequently, the negative result for  $\gamma$  is consistent with the diversity hypothesis.



The results can be shown by graphically presenting the functional relationships of the theoretical model and applying the parameter estimates.<sup>17</sup> Figure 1 visualises the relationship between  $S$  and  $ELF$  and, respectively,  $\bar{D}$  and  $TFP$  according to equations (4) and (5), for the GMM-SYS estimates and assuming an initial level of  $TFP = 1$ . From the figure, it follows that both  $S$  and  $ELF$  may positively contribute to productive diversity ( $\bar{D}$ ). Corner values in Figure 1 are informative in this respect. By fixing  $ELF$  at 0 (that is, no ethnic diversity within the immigrant workforce), any increase in  $S$  induces an increase of  $\bar{D}$  and, therefore, of  $TFP$ . The highest value in Figure 1 is reached for  $ELF = 0$  and  $S = 0.8$ : with no ethnic diversity, 80 per cent of immigrants on the total workforce implies 0.7 of productive diversity and a 49 per cent increase in productivity. A similar outcome, but quantitatively less realistic, emerges for  $ELF$  when  $S$  is fixed at a low value. For instance, for  $S = 0.05$  (i.e. a 5 per cent presence of immigrants) an increase of diversity always produces an increase in  $\bar{D}$  and  $TFP$ : at  $ELF = 0.5$  (a remarkable ethnic diversity within the immigrant workforce) productive diversity reaches 0.08 with a consequent 4.5 per cent  $TFP$  growth.

Figure 1 also shows that the monotonic positive impact of both  $S$  and  $ELF$  on productivity disappears for higher values of both these terms and, above all, when both increase simultaneously. Ultimately, for large enough values of  $S$  and  $ELF$  their combined contribution to productive diversity becomes negative, that is, generates unproductive diversity and a  $TFP$  decline. For instance, when  $ELF = 0.58$ , an increase of  $S$  produces a positive impact on  $\bar{D}$  and  $TFP$  up to  $S = 0.5$ . Then, for a share of immigrants higher than 50 per cent, any increase of  $S$  implies a negative impact on productivity. At the same time, for a 10 per cent share of immigrants, an increase of  $ELF$  produces a positive impact on  $\bar{D}$  and  $TFP$  up to  $ELF = 0.09$ . Beyond this level, any increase of  $ELF$  implies a negative impact on productivity. It is worth noticing that the average values of  $S$  and  $ELF$  within the adopted panel (<5 per cent and 0.05, respectively) indicate that the Italian farm sample under consideration is still largely operating in the space where the combination of  $S$  and  $ELF$  produces a positive impact on productive diversity and, therefore, on  $TFP$ .

In Figure 1, the shaded areas indicate  $S$ – $ELF$  combinations implying the same (range of) values of  $\bar{D}$  and  $TFP$ . The lightest grey area indicates the highest values of  $\bar{D}$  and  $TFP$ , and it is positioned in the upper left corner where both  $S$  and  $ELF$  assume low values. The darkest area indicates all combinations of  $S$  and  $ELF$  with a negative impact on  $\bar{D}$  and  $TFP$  (i.e.  $TFP < 1$ ). This area is positioned in the lower right corner with a minimum value of  $\bar{D} = -0.18$  and  $TFP = 0.91$  reached when both  $S$  and  $ELF$  equal 0.8.

Figure 1 reveals how the model estimates provide empirical support for the diversity hypothesis. This can be made more explicit by showing how these

17 In drawing Figures 1 and 2, the lagged values of  $S$  and  $ELF$ , thus terms  $(\theta_{Lit-1}ELF_{it-1} - \theta_{Lit-1}S_{it-1})$  and  $(\theta_{Lit-1}S_{it-1}ELF_{it-1})$ , have been disregarded as the estimates of the respective coefficients are not statistically different from 0, thus confirming what obtained in the estimation of the unrestricted model.

estimates lead to the inverted-U shape often mentioned in the extant literature on this topic. According to equation (6) and assuming an initial level of  $TFP = 1$ , the estimated  $\gamma$  allows, for a given value of the labour share ( $\theta_{Lit}$ ), to draw the functional relationship between  $TFP$  and  $(S + ELF)$ .<sup>18</sup> Figure 2a firstly displays this functional relationship according to the GMM-SYS estimates with the corresponding 95 per cent confidence bounds. Figure 2b displays this function under four possible values of  $\theta_{Lit}$  (0.1; 0.5; 0.9; the sample average, that is, 0.514) and the two GMM estimates of  $\gamma$  (-1.476 and -1.532, respectively). An inverted U-shape clearly emerges thus confirming the validity of the diversity hypothesis.

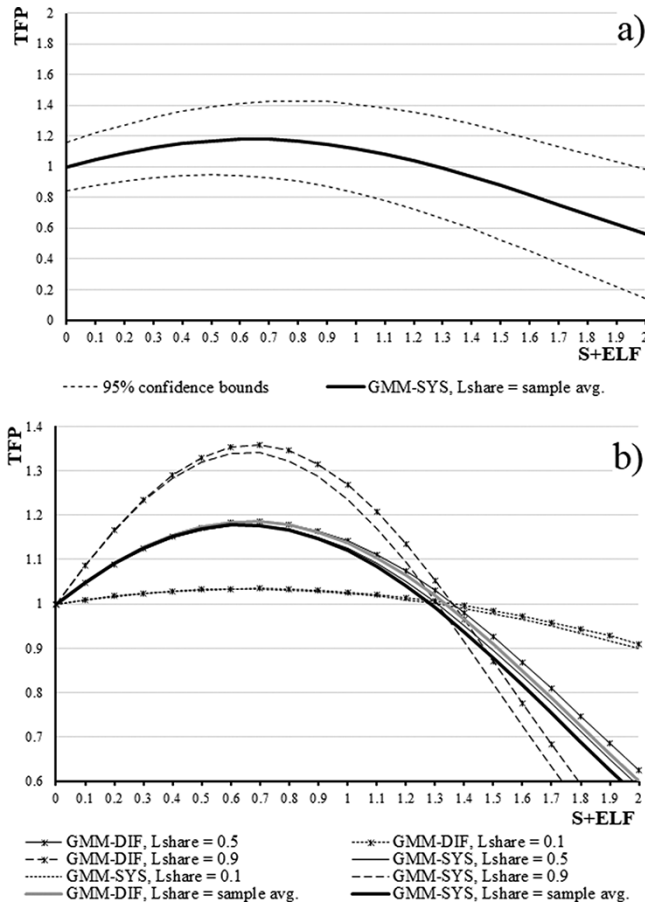
Two aspects are worth noticing about this inverted-U shape. Firstly, the shape of this non-monotonic relationship is mostly affected, as expected, by the labour share, which is, according to the theoretical derivation, the medium of the impact of diversity (thus, of immigrant workforce) on productivity. This explains why with a low labour share, the impact of diversity is constantly low, while for a high labour share, this impact can be large but also largely variable, depending on how large the diversity is. Secondly, the turning point is observed at a quite large value of  $(S + ELF)$ . For the GMM-SYS estimates and the observed labour share (Figure 2a), the combination of the presence and composition of the immigrant workforce starts having a negative impact on farm-level productivity at  $(S + ELF) = 0.75$ . For the average ethnic diversity observed within the panel sample ( $ELF = 0.05$ ), this means that the turning occurs at  $S = 0.7$ , that is for a very large presence of immigrants. Such a share (70 per cent) is almost 25 times higher than the average share observed for the Italian farm panel. Therefore, this turning point is more a theoretical hypothesis than a realistic perspective as these critical values of  $S$  and  $ELF$  are far from the levels observed. If we applied this result to the current presence of immigrants in Italian agriculture, the implication would be that a negative impact on  $TFP$  (given  $ELF = 0.05$ ) would occur only by increasing the presence to almost 1 million units.

### 5.3. Comparing estimators

Tables 2 and 3 show that the GMM-DIF and GMM-SYS estimates are similar. Compared with the GMM-DIF results, the GMM-SYS estimates yield a small gain in terms of estimation efficiency particularly for exogenous variables  $ES$  and those expressing production specialisation. But this gain is not obvious in the present case as the autocorrelation coefficient of the dependent variable is far from being close to 1, that is, when the advantage of performing GMM-SYS rather than GMM-DIF estimation is expected to be larger (Arellano, 2003).

The usual tests performed on these GMM estimations (bottom of Tables 2 and 3) corroborate the reliability of the GMM-SYS results. AR(1) and AR(2)

18 A similar relationship could be also obtained using weights  $w$  and  $(1-w)$ , as they appear in the functional relationship linking  $TFP$  to  $S$  and  $ELF$  (equation (5)). Nonetheless, as evident in equation (6), the variation of  $TFP$  does not depend on these weights. Moreover, in the adopted form, Figure 2 is of more immediate interpretation: by fixing the value of  $ELF$  ( $S$ ), any unit movement on the horizontal axis can be easily interpreted as an unit variation of  $S$  ( $ELF$ ) with the consequent response of  $TFP$  on the vertical axis.



**Fig. 2.** Relationship between the farm-level labour force diversity ( $S + ELF$ ) and productivity ( $TFP$ ). Notes: The functional relationships are obtained according to equation (6). Figure 2a displays the 95 per cent confidence bounds for the GMM-SYS estimation (with the sample average of the labour share). Figure 2b displays the functional relationship under three possible values of labour share ( $\theta_{Lit}$ ) and the two GMM estimates of the  $S-ELF$  interaction parameter ( $\gamma$ ).

tests confirm that the dynamic model is properly specified (Arellano, 2003).<sup>19</sup> The Sargan test of instrument selection is not passed at the 10 per cent and 5 per cent significance level for the GMM-SYS estimation (unrestricted model) and GMM-DIF estimation (restricted model), respectively. However, the Sargan test is reliable only under homoscedastic errors, an assumption that is hardly tenable in panel data sets like the one used here. To confirm the instruments' validity, the Hansen test is also reported as it generalises the Sargan test, together with the difference-in-Hansen test for the GMM-SYS estimation,

19 For the lagged dependent variable in (11)–(12), all available and admitted lagged  $TFP$  levels are considered as instruments in the GMM-DIF estimation then augmented with all available and admitted lagged first-differenced  $TFP$  levels in the GMM-SYS estimation.

that checks the validity of a subset of instruments and thus is particularly helpful in the GMM-SYS estimation as it uses instruments for the levels equation, in addition to the instruments used in the GMM-DIF estimation. All these additional tests confirm the validity of the adopted instruments, as the null hypothesis assumes that all the selected instruments are valid, that is, exogenous.

The estimated coefficients of the time (year) dummies are not reported in both [Tables 2](#) and [3](#). These estimates are available upon request.

## 6. Concluding remarks and policy implications

This paper investigates the farm-level relationship between TFP and the presence and composition of immigrant workforce. It proposes a modelling framework that explains the complex impact of immigrant workers on farm-level productivity. This model leads to a testable specification of the complex relationship, commonly referred to as the ‘diversity hypothesis’. Secondly, the paper proposes an empirical strategy to assess this hypothesis and presents an application using a balanced panel of Italian professional farms extracted from the FADN over the period 2008–2015. The use of this micro-level data enables the wide heterogeneity across farms to be considered. The results confirm the diversity hypothesis: immigrant workforce contributes to higher agricultural productivity provided that its share of the workforce is not too high and/or its composition is homogeneous. The results further show the complexity of this effect, which consists of a non-monotonic response to the presence of immigrant workers and the composition of this workforce in combination. A possible negative impact would require values for the presence and composition of the immigrant workforce that are currently unrealistic.

The conclusion comes with a caveat. Caution is needed in generalising the results as they refer to a limited time-period (2008–2015) and to a specific sector and country (Italian agriculture). Therefore, further analysis for other periods and production contexts (farm samples, sectors and countries), as well as a further development of the theoretical background, is needed to validate the empirical evidence. Moreover, it remains true that a deeper understanding of the role of diversity and the farm-level TFP performance could be obtained by specifying and estimating a stochastic production frontier. However, several empirical issues remain to be addressed for the situation of large farm heterogeneity, and further research is necessary.

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## Supplementary data

Supplementary data are available at *ERA* online.

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